

Nordtronics: laptop builds, VPS serves

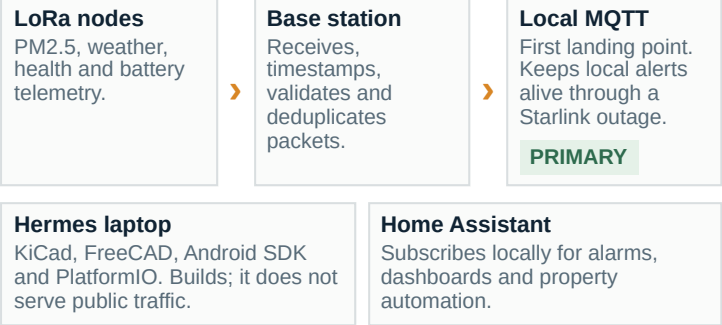
A small always-on cloud half for the main site, fire data, webhooks, and the future app.

about \$5–6/mo

Planning target for a Hetzner-class shared VPS. Backups, domain, storage, taxes, or IPv4 can add to it.

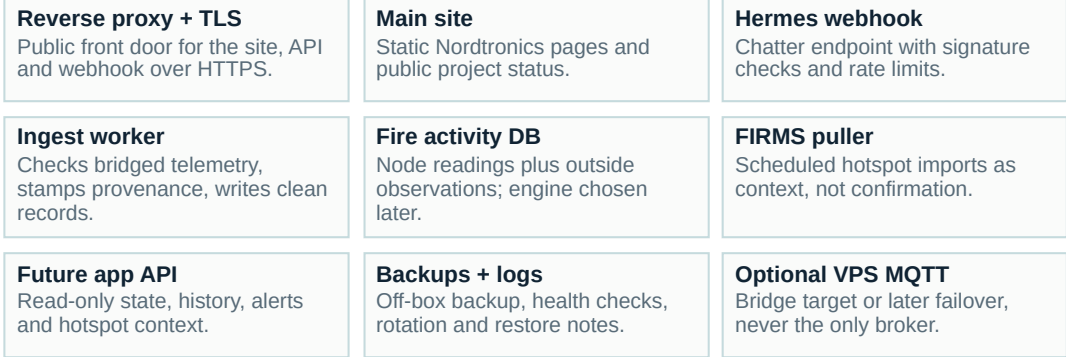
Home / edge: hardware stays close

Desktop tools and local control live here.



VPS: the always-on public half

One small server to start; split services only when traffic proves it necessary.



The data path stays simple

1. **Node** → **base station**: send a compact reading over LoRa.
2. **Base station** → **local MQTT**: timestamp, validate, deduplicate, then publish.
3. **Local MQTT** → **VPS**: bridge selected topics over Transport Layer Security (TLS), with retry buffering.
4. **Ingest** → **database**: validate again, stamp provenance, and write telemetry.
5. **Public feeds** → **database**: a separate job imports NASA FIRMS hotspot observations.
6. **Database** → **clients**: the website and future app API read a stable model, never the field nodes.

Starlink down?

Collection and Home Assistant continue locally. Cloud views go stale; queued telemetry catches up later.

Build it in this order

1. **Harden one small VPS.**
SSH keys, firewall, updates, TLS, monitoring, and off-box backup.
2. **Host the static site.**
Prove DNS, deploy, certificates, backup, and rollback.
3. **Add the Hermes webhook.**
One narrow job; signed requests, short logs, hard rate limit.
4. **Bridge one real node.**
One topic schema, durable buffering, timestamps in UTC.
5. **Add storage, then FIRMS.**
Choose storage from real queries; add outside context second.
6. **Add the app API last.**
Build it when an actual client and stable read model exist.

Decisions to leave open

Database engine

Plain PostgreSQL, a time-series extension, or a dedicated store? Decide from real queries.

Retention

Raw-data lifetime, downsampling and durable alert evidence.

Region

Colorado latency, price, data location and backup geography.

API boundary

Public map data, owner data, or both? Never expose raw MQTT topics.

Cloud MQTT

Bridge target first; failover only when a real need appears.

Do not build ahead of the hardware.

No rushed database, full app API, Kubernetes or multi-server layout. One box. Good backups.